

Investigative Report: SQL Murder Mystery Analysis

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Tool: PostgreSQL (pgAdmin 4)

Database: SQL City Police Department Database

1. Introduction

This report documents a structured SQL-based criminal investigation conducted using the SQL City Police Department database. The objective was to identify the perpetrator of a murder that occurred on January 15, 2018, in SQL City and subsequently determine the individual who orchestrated the crime. Through a systematic process involving data retrieval, filtering, table joins, and investigative analysis, the investigation successfully identified Jeremy Bowers as the murderer and Miranda Priestly as the mastermind behind the crime. Although presented as a fictional investigation, this project mirrors real-world applications of SQL in:

- Digital forensics
- Law enforcement intelligence
- Fraud detection
- Insurance investigations
- Compliance and audit analytics
- Relational database querying

The investigation demonstrates how structured query language transforms fragmented records into actionable intelligence through logical filtering and relational analysis.

Background

The SQL City Police Department reported a murder that occurred on January 15, 2018, within SQL City. The original crime scene report had been misplaced, requiring investigators to reconstruct the case using information stored within the police database.

Objective

The objectives of this investigation were:

1. Retrieve the crime scene report.
2. Identify and interview witnesses.
3. Determine the murder suspect.
4. Verify the suspect using supporting evidence.
5. Identify the individual who commissioned the murder.

Database Overview

The investigation relied on several interconnected tables within the SQL City Police Department database.

Table	Purpose
crime_scene_report	Crime incident records
Person	Personal identification and addresses
Interview	Witness and suspect statements
get_fit_now_member	Gym membership records
get_fit_now_check_in	Gym attendance logs
drivers_license	Vehicle and physical descriptions
facebook_event_checkin	Event attendance history
Income	Financial records

The relational nature of these tables enabled evidence to be linked through unique identifiers such as `person_id`, `license_id`, and `membership_id`.

2. Investigation Methodology

The investigation followed a structured data analysis approach:

1. Crime scene report retrieval.
2. Witness identification.
3. Witness interview analysis.
4. Suspect identification through gym membership records.
5. Verification using gym attendance records.
6. Vehicle registration analysis.
7. Suspect confirmation through table joins.
8. Interrogation transcript analysis.
9. Identification of the mastermind through event attendance and demographic profiling.

3. Crime Scene Report Analysis

Query Executed

The first step involved retrieving the crime scene report matching the known details:

- Crime Type: Murder
- Date: January 15, 2018
- Location: SQL City

```
SELECT *  
FROM crime_scene_report  
WHERE crime_type = 'murder'  
AND date = 20180115  
AND city = 'SQL City';
```

Findings

The report revealed that:

- Two witnesses observed critical details related to the crime.
- The witnesses were located using address-based clues provided in the report.

4. Witness Identification

Witness 1: Last House on Northwestern Drive

Query:

```
SELECT *  
FROM person  
WHERE address_street_name = 'Northwestern Dr'  
ORDER BY address_number DESC;
```

Result

Name	ID
Morty Schapiro	14887

Witness 2: Annabel on Franklin Avenue

Query:

```
SELECT *  
FROM person  
WHERE name LIKE 'Annabel%'  
AND address_street_name = 'Franklin Ave';
```

Result

Name	ID
Annabel Miller	16371

5. Witness Interview

Interview Retrieval

```
SELECT *  
FROM interview  
WHERE person_id IN (14887,16371);
```

Key Statements

Morty Schapiro

Reported:

- Heard a gunshot.
- Saw a man fleeing the scene.
- The individual carried a **Get Fit Now Gym bag**.
- Membership number began with "**48Z**".
- The bag was exclusive to **Gold Members**.
- The suspect drove a vehicle with plate number containing "**H42W**".

Annabel Miller

Reported:

- Personally witnessed the murder.
- Recognized the suspect from **Get Fit Now Gym**.
- Saw him at the gym on **January 9, 2018**.

6. Suspect Identification

Query

```
SELECT *
FROM get_fit_now_member
WHERE id LIKE '48Z%'
AND membership_status = 'gold';
```

Results

Suspect	Membership ID	Person ID
Joe Germuska	48Z7A	28819
Jeremy Bowers	48Z55	67318

A third member, Tomas Baisley (48Z38), was excluded because he held a Silver membership, contradicting witness testimony.

7. Verification Using Gym Check-In Records

Query

```
SELECT *
FROM get_fit_now_check_in
WHERE membership_id IN ('48Z7A', '48Z55', '48Z38')
AND check_in_date = 20180109;
```

Findings

Name	Check-in Date
Joe Germuska	Present
Jeremy Bowers	Present
Tomas Baisley	Not Present

This confirmed that only Joe Germuska and Jeremy Bowers matched both witness statements.

8. Vehicle Registration Analysis

Witness Vehicle Description

The suspect's license plate contained:

H42W

Query

```
SELECT *  
FROM drivers_license  
WHERE plate_number LIKE '%H42W%'  
AND gender = 'male';
```

Results

License ID	Plate Number
423327	0H42W2
664760	4H42WR

9. Suspect Confirmation

To associate vehicle records with individuals, a join was performed between the drivers_license and person tables.

Query

```
SELECT  
dl.age,  
dl.height,  
dl.hair_color,
```

```
dl. eye_color,  
dl. gender,  
dl. plate_number,  
dl. car_make,  
dl. car_model,  
p. name,  
p. id  
FROM drivers_license dl  
LEFT JOIN person p  
ON dl. id = p. license_id;
```

Matching Results

Cross-referencing gym membership records and vehicle ownership revealed:

Name	Membership ID	Plate Number
Jeremy Bowers	48Z55	0H42W2

Conclusion

All evidence aligned with **Jeremy Bowers**, confirming him as the murderer.

Phase Two: Identifying the Mastermind

10. Interrogation of Jeremy Bowers

Query

```
SELECT *  
FROM interview  
WHERE person_id = 67318;
```

Jeremy's Statement

Jeremy admitted that:

- He was hired to commit the murder.
- The employer was a wealthy woman.
- She had:
 - Red hair.
 - Height between 65 and 67 inches.
 - Drove a Tesla Model S.
- She attended the [SQL Symphony Concert](#) three times in [December 2017](#).

11. Suspect Profiling

Query

```
SELECT *  
FROM full_details  
WHERE height BETWEEN 65 AND 67  
AND hair_color = 'red'  
AND car_make = 'Tesla'  
AND car_model = 'Model S'  
AND gender = 'female';
```

Results

Name	Person ID
Red Korb	78881
Regina George	90700
Miranda Priestly	99716

12. Event Attendance Verification

Query

```
SELECT *  
FROM facebook_event_checkin  
WHERE person_id IN ('78881', '90700', '99716');
```

Findings

Only one suspect attended the [SQL Symphony Concert](#) exactly three times in [December 2017](#):

Name	Attendance Dates
Miranda Priestly	Dec 6, Dec 12, Dec 29

13. Financial Validation

Income records further confirmed Miranda Priestly's profile as a wealthy individual.

Query

```
SELECT *  
FROM income  
WHERE annual_income >= 310000;
```

Result

Miranda Priestly met the financial profile described by Jeremy.

Final Findings

Murderer Identified

Attribute	Value
Name	Jeremy Bowers
Person ID	67318
Gym Membership	48Z55
Vehicle Plate	0H42W2

Jeremy Bowers matched all witness descriptions, gym records, and vehicle evidence, conclusively identifying him as the individual who committed the murder.

Mastermind Identified

Attribute	Value
Name	Miranda Priestly
Person ID	99716
Vehicle	Tesla Model S
Hair Color	Red
Height	65–67 inches
Concert Attendance	3 times in December 2017

Miranda Priestly matched every characteristic provided by Jeremy Bowers and was identified as the individual who orchestrated the murder.

Conclusion

Through the application of SQL querying techniques including filtering, pattern matching, record linkage, joins, and behavioral verification, the investigation successfully reconstructed the murder case. Evidence gathered from witness testimony, gym membership records, attendance logs, vehicle registrations, interview transcripts, event participation records, and income data led to the identification of:

- **Jeremy Bowers** as the murderer.
- **Miranda Priestly** as the individual who commissioned the crime.

This investigation demonstrates the effectiveness of relational database analysis in solving complex investigative cases and highlights how SQL can be used as a powerful tool for forensic data analysis and evidence-based decision-making.